

BLACKFIN

Flashing Procedure Guide

Issue 1.0

Document History

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1. INTRODUCTION

The purpose of this document is to explain the flashing procedure of BLACKFIN in EVAL-Melody-8 Board.

2. HARDWARE REQUIREMENTS

- EVAL-Melody-8 board
- Power adapter
- RS232 (Female-to-male end)
- PC

3. SYSTEM REQUIREMENTS

- Windows 98/2000/XP/07/10 installed.
- Tera Term 4.60

4. APPLICATION SOFTWARE INSTALLATION

4.1. BLACKFIN FLASHING PROCEDURE

- Install teraterm-4.60.exe.
- Unzip the application firmware “*Eval_Melody_8_Blackfin_Micro.bin*” in the local.
- Connect the RS232 cable from PC to Board UART port as show below.

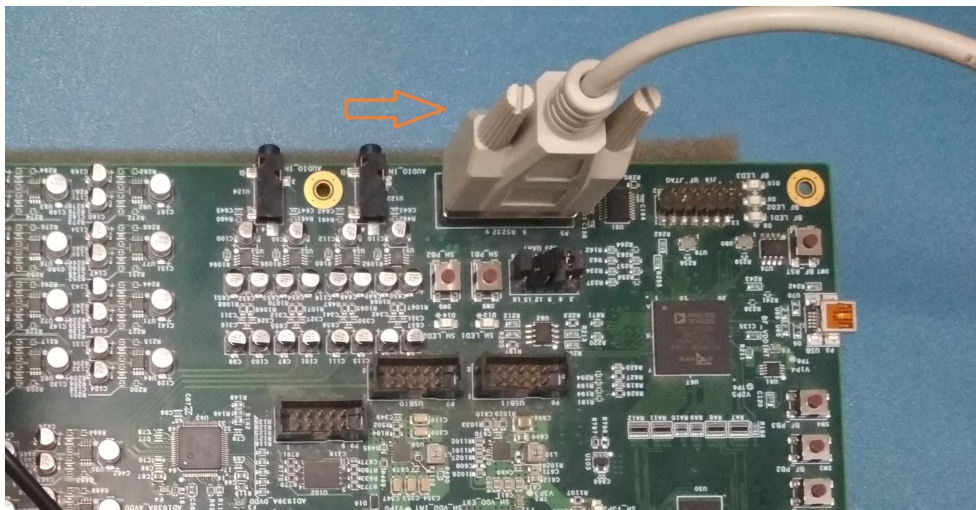


Figure 1:UART cable connected with serial port

- Open Tera Term 4.60, select serial port set the below settings.

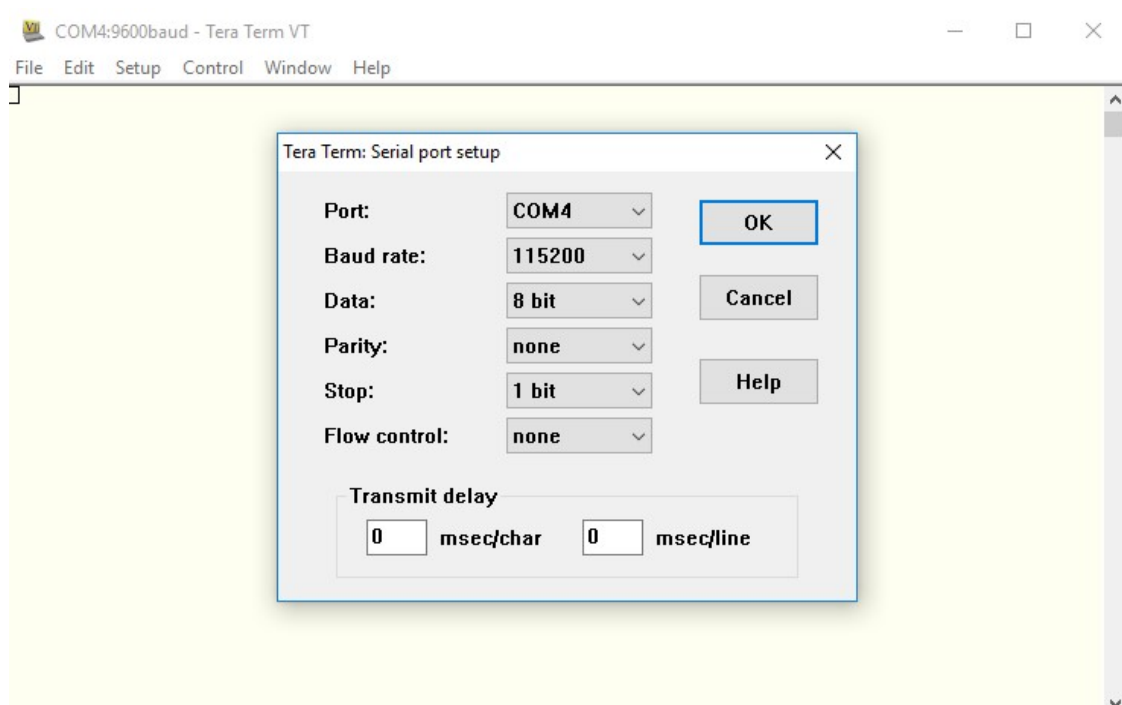


Figure 2: Tera term 4.60 Window

- Power on the EVAL-Melody-8 board.
- The following message appears in Tera Term window

U-Boot 2013.07 (ADI-2014R1) (Feb 22 2018 - 12:54:08)

CPU: ADSP bf524-0.2 (Detected Rev: 0.2) (spi flash boot)

Board: ADI BF526 EZ-Board board

Support: <http://blackfin.uclinux.org/>

Clock: VCO: 300 MHz, Core: 300 MHz, System: 100 MHz

RAM: 32 MiB

SF: Detected MX25L3205D with page size 64 KiB, total 4 MiB

In: serial

Out: serial

Err: serial

KGDB: [on serial] ready

Hit any key to stop autoboot

- After display “Hit any key to stop autoboot” and count in prompt, press any key during the countdown. Then the prompt will display the following,
bf>

- At the prompt, type the following command “*sf probe 0:I*” and press enter. Then the prompt will display the following,
SF: Detected MX25L3205D with page size 64 KiB, total 4 MiB
bfin>
- At the prompt, type the following command “*sf erase 0x60000 0xa0000*” and press enter. Then the prompt will display the following,
bfin>
- At the prompt, type the following command “*loady*” and press enter. Then the prompt will display the following,
Ready for binary (ymodem) download to 0x00100000 at 115200 bps...
C
- In Tera Term, under **File**, click **Transfer**, then select **YMODEM or XMODEM** and Send.
Select the application firmware (*Eval_Melody_8_Blackfin_Micro.bin*) and click Open.
- After the transfer is complete, then the prompt will display the following,
CCxyzModem - CRC mode, 1969(SOH)/0(STX)/0(CAN) packets, 6 retries
Total Size = 0x0003d7d4 = 251860 Bytes
bfin>
- At the prompt, type the following command to program the application area of the SPI flash memory:
“sf write \$(loadaddr) 0x60000 \$(filesize)”
- Then the prompt will display the following,
bfin>
- Reset the board and allow the countdown to complete. Now U-Boot should launch the application firmware that just programmed.
- Now Blackfin LED blinking, which indicates Blackfin enable successfully as shown in the figure 3.

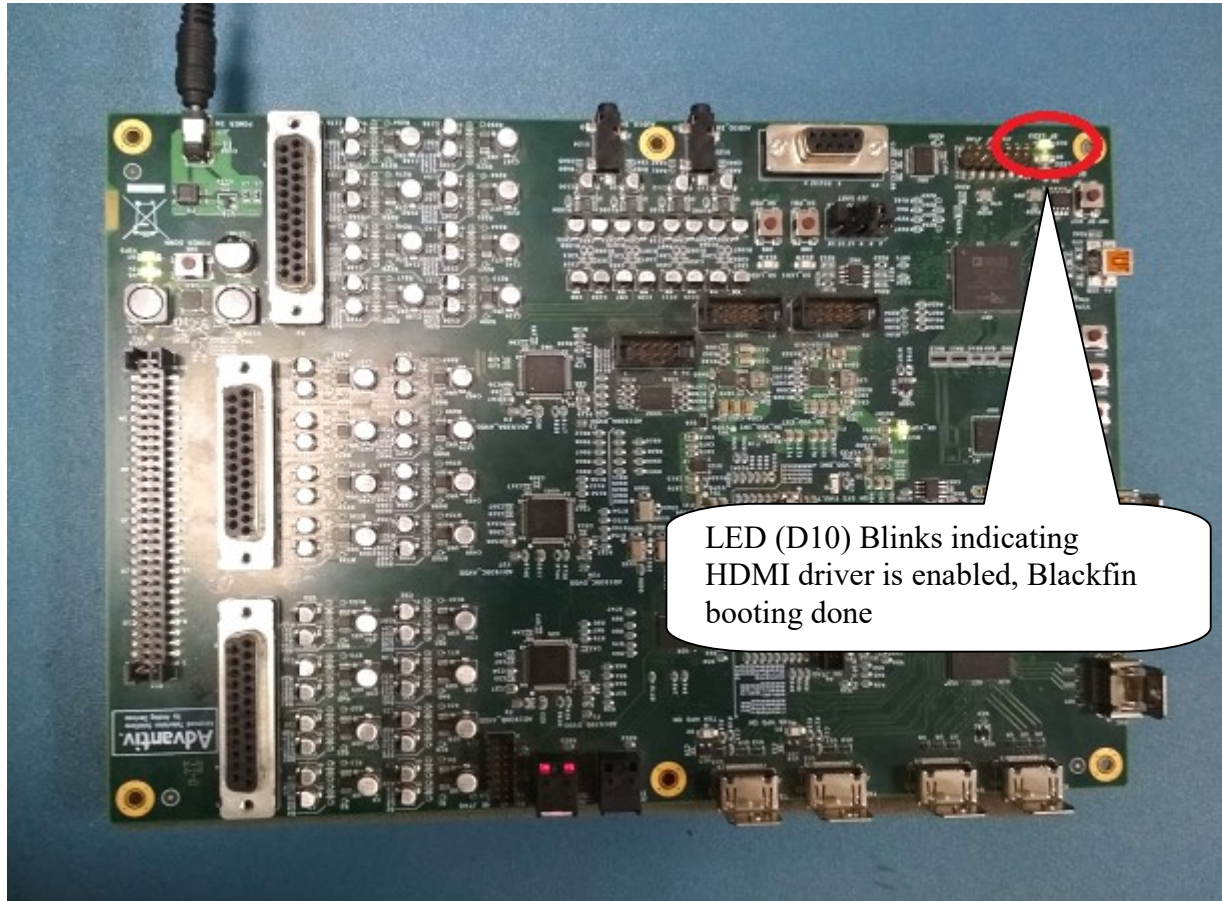


Figure 3: Blackfin boot indication
